

Problem Statement or Requirement:

A client's requirement is, he wants to predict the insurance charges based on the several parameters. The Client has provided the dataset of the same.

As a data scientist, you must develop a model which will predict the insurance charges.

1.) Identify your problem statement: Insurance charge prediction based on personal information

2.) Tell basic info about the dataset (Total number of rows, columns) .- Dataset is having 6 columns (age, sex, bmi, children, smoker, charges) and 1337 rows of data

3.) Mention the pre-processing method if you're doing any (like converting string to number – nominal data)

Given dataset is having 'Nominal data' for 'sex' and 'smoker' column which needs to be converted into numerical data before passing to any algorithm as it will not process the categorical data. Here we are using 'one Hot encoding' method using 'get_dummies' method from 'pandas' library.

Given Dataset:

age	sex	bmi	children	smoker	charges
19	female	27.900	0	yes	16884.92400
18	male	33.770	1	no	1725.55230
28	male	33.000	3	no	4449.46200
33	male	22.705	0	no	21984.47061

Pre-processed dataset:

age	bmi	children	charges	sex_male	smoker_yes
19	27.900	0	16884.92400	False	True
18	33.770	1	1725.55230	True	False
28	33.000	3	4449.46200	True	False
33	22.705	0	21984.47061	True	False

4.) Develop a good model with r2_score. You can use any machine learning algorithm; you can create many models. Finally, you have to come up with final model.

Tried the below 4 algorithm for the given dataset and found the high R2 value in 'Random forest'.

1. Multiple Linear
2. Support Vector Machine
3. Decision Tree
4. Random Forest

5.) All the research values (r2_score of the models) should be documented. (You can make tabulation or screenshot of the results.)

Please find the below R2 value for 4 different algorithm.

1. Multiple Linear Regression [R2 value = 0.7894790349867009]

```
: LinearRegression ⓘ ⓘ  
  ▶ Parameters  
  
: y_pred=regressor.predict(X_test)  
  
: from sklearn.metrics import r2_score  
  r2_value=r2_score(Y_test,y_pred)  
  r2_value  
  
: 0.7894790349867009
```

2. Support vector machine (SVM)

S.No	Hyper parameter C	kernel	R2_value
1	0.1	Linear	-0.122076684
2	6	Linear	-0.049888616
3	20	Linear	0.113310171
4	50	Linear	0.54328182
5	500	Linear	0.627046276
6	1000	Linear	0.634036931
7	15000	Linear	0.744504776
8	15500	Linear	0.742139967
9	20000	Linear	0.741960677
10	0.1	rbf	-0.089576246
11	10	rbf	-0.081969104
12	12	rbf	-0.08065991
13	12	poly	-0.093756009
14	0.368685	poly	-0.089224597

15	0.368685	sigmoid	-0.089836289
16	100	sigmoid	-0.118145548
17	7.89	sigmoid	-0.09028

Support Vector Machine R2_value is high for **kernel='Linear'** , **C=15000**, **R2= 0.744504776**

3. Decision Tree

S.No	criterion	splitter	max_features	r2_SCORE
1	squared_error	best	None	0.705790579
2	squared_error	best	1	0.707774023
3	squared_error	best	0.5	0.745370799
4	squared_error	best	sqrt	0.70345308
5	squared_error	best	log2	0.67642251
6	squared_error	random	None	0.752797562
7	squared_error	random	1500	0.712749267
8	squared_error	random	0.99	0.742155281
9	squared_error	random	sqrt	0.706425202
10	squared_error	random	log2	0.685887626
11	absolute_error	best	None	0.667956301
12	absolute_error	best	60	0.66965545
13	absolute_error	best	0.99	0.711021735
14	absolute_error	best	sqrt	0.611214401
15	absolute_error	best	log2	0.654817037
16	absolute_error	random	None	0.738806726
17	absolute_error	random	1	0.691930954
18	absolute_error	random	0.7	0.792107224
19	absolute_error	random	sqrt	0.661675339
20	absolute_error	random	log2	0.738518102
21	poisson	best	None	0.723362748
22	poisson	best	127	0.736342693
23	poisson	best	0.75	0.75377107
24	poisson	best	sqrt	0.712395449
25	poisson	best	log2	0.696533883
26	poisson	random	None	0.727665107
27	poisson	random	95	0.77359916
28	poisson	random	0.97	0.725293615
29	poisson	random	sqrt	0.551946333
30	poisson	random	log2	0.655581472

Decision tree R2 value is high for **criterion= absolute_error**, **splitter= random** , **max_features= 0.7=> R2_value is 0.792107223596603**

4. Random forest

Random forest R2 value is high for **n_estimators = 100** ,**random_state=0**=> R2_value is **0.8538307913484513**

```
from sklearn.ensemble import RandomForestRegressor
regressor=RandomForestRegressor(n_estimators=100,random_state=0)
regressor.fit(X_train,Y_train)
```

C:\Users\subas\anaconda3\Lib\site-packages\sklearn\base.py:1365: DataC
se change the shape of y to (n_samples,), for example using ravel().
return fit_method(estimator, *args, **kwargs)

▼ RandomForestRegressor ⓘ ?

► Parameters

```
y_pred=regressor.predict(X_test)
```

```
from sklearn.metrics import r2_score
r2_value=r2_score(Y_test,y_pred)
r2_value
```

#estimator- 10 - 0.83303041340085
150 - 0.8531705916190235
#100-0.8538307913484513

0.8538307913484513

6.) Mention your final model, justify why u have chosen the same.

Best model for the given dataset is 'Random Forest model'. Because R2 value is high (0.85383) this model compared to others. Hence saving this model as final model and using it for deployment.

```

return fit_method(estimator, *args, **kwargs)
8]: RandomForestRegressor ⓘ ⓘ
    Parameters

9]: y_pred=regressor.predict(X_test)

0]: from sklearn.metrics import r2_score
    r2_value=r2_score(Y_test,y_pred)
    r2_value

    #estimator- 10 - 0.83303041340085
    # 150 - 0.8531705916190235
    #100-0.8538307913484513

0]: 0.8538307913484513

1]: import pickle
    modelName = 'regression_assign_randomForest_final.sav'
    pickle.dump(regressor,open(modelName,'wb'))

]:

```

Deployment and real data validation

```

]: import pickle

]: loaded_model = pickle.load(open('regression_assign_randomForest_final.sav','rb'))
   result=loaded_model.predict([[25,29,2,0,0]])
   result

C:\Users\subas\anaconda3\Lib\site-packages\sklearn\utils\validation.py:2749: UserWarning:
as fitted with feature names
  warnings.warn(

]: array([4476.2216785])

```